

1. Administrative & Study Identification

Full Study Title: A Phase 3, Randomized, Double-blind, Placebo-controlled, Study to Evaluate the Efficacy and Safety of Adjunctive KarXT for the Treatment of Mania, With or Without Mixed Features, in Individuals With Bipolar-I Disorder Taking Lithium, Valproate, or Lamotrigine **Short Title/Acronym:** Adjunctive KarXT for the Treatment of Mania in Bipolar-I Disorder **Protocol Number:** CN012-0046 **NCT Number:** NCT07140913 **Posting ID:** 173442267910824732 **Sponsor/Company:** Bristol Myers Squibb **Investigational Product:** KarXT (Xanomeline/Trospium Chloride) **Phase of Development:** Phase 3 (Phase Requirement: Trial must be in PHASE3) **Trial Status:** Recruiting (Currently recruiting participants) **Source Level:** 5 **Medical Monitor:** BMS Clinical Trials Contact Center

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the efficacy and safety of adjunctive KarXT for the treatment of mania, with or without mixed features, in participants with Bipolar-I Disorder taking lithium, valproate, or lamotrigine. **Secondary Objectives:** To evaluate improvements in illness severity, global clinical impressions (e.g., CGI-BP), and assess short-term adverse events and tolerability profiles.

2.2 Background & Rationale To address the need for more effective and tolerable treatments for bipolar mania. Unlike standard treatments for bipolar disorder, which often rely solely on mood stabilizers, KarXT combines xanomeline, a muscarinic receptor agonist, with trospium chloride, which reduces unwanted side effects. This unique mechanism targets both the M1 and M4 muscarinic receptors in the brain, potentially improving cognition and mood symptoms more effectively and with a different side-effect profile than current options.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled **Blinding:** Double-blind **Randomization:** Randomized

3.2 Planned Enrollment & Duration Targeted Enrollment (\$N\$): 424 participants **Number of Sites:** Global multicenter study (including US, Europe, Japan, China, Argentina, and Rest of World) **Estimated**

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age 18 to 65 years with a BMI ≥ 18 and ≤ 40 kg/m². 2. Primary diagnosis of Bipolar-I disorder established by a comprehensive psychiatric evaluation based on the DSM-5-TR criteria and confirmed by the Mini International Neuropsychiatric Interview (MINI) version 7.0.2 Standard. 3. Individual is experiencing an acute exacerbation or relapse of a manic episode, with or without mixed features (≤ 3 weeks), requiring hospitalization. 4. Currently experiencing an acute episode of mania or mania with mixed features with a therapeutic dose of lithium, valproate, or lamotrigine. The dose of the mood stabilizer must have remained stable for at least two weeks prior to screening. Participants on valproate must have been receiving treatment with valproate for a minimum of seven months. 5. Young Mania Rating Scale (YMRS) Total Score of ≥ 18 at Screening and at Baseline, and $< 20\%$ reduction in YMRS from screening to baseline. 6. Clinical Global Impression Severity scale (CGI-BP) ≥ 4 .

4.2 Exclusion Criteria 1. History of irritable bowel syndrome (with or without constipation) or any serious constipation requiring treatment within the last 6 months. 2. History or high risk of urinary retention, gastric retention, or narrow-angle glaucoma. 3. Participants with HIV, cirrhosis, biliary duct abnormalities, hepatobiliary carcinoma, and/or active hepatic viral infections based on either medical history or the LFT results. 4. Elevations in hepatic transaminases at screening $\geq 2 \times$ ULN for ALT and AST and/or bilirubin $> 1.5 \times$ ULN, unless in the context of Gilbert's syndrome. All grades of hepatic impairment (mild [Child-Pugh Class A], moderate [Child-Pugh Class B], and severe [Child-Pugh Class C]). 5. Any primary DSM-5-TR disorder other than BP-I within 12 months before screening (confirmed using MINI version 7.0.2 Standard) including BP-I depression, BP-I with rapid cycling, first manic episode, BP-II, and major depressive disorder. 6. Individual has a DSM-5-TR diagnosis of moderate to severe substance use disorder (except tobacco use disorder) within the 12 months before screening, or current use as determined by urine toxicology screen or alcohol test. 7. Risk for suicidal behavior at screening as determined by the investigator's clinical assessment and the C-SSRS with an answer "Yes" to item 4 or 5 within 6 months before screening or between screening and baseline, or suicide attempt within 12 months before screening, or between screening and baseline.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Adjunctive KarXT (Xanomeline/Trospium Chloride) administered orally alongside the patient's background mood stabilizer (e.g., Lithium). **Route of Administration:** Oral **Duration of Treatment:** Acute treatment phase during hospitalization, followed by safety extension and monitoring.

5.2 Concomitant & Prohibited Therapy Permitted: Stable background therapeutic doses of lithium (Drug Preferred Name: lithium, ATC Code: N05AN01), Valproate, or Lamotrigine. **Prohibited:** Other non-study antipsychotics or treatments that could confound efficacy, or medications known to exacerbate anticholinergic or cholinergic risks.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -14 to 0	Informed Consent, DSM-5-TR/MINI 7.0.2 Psychiatric Evaluation, YMRS baseline assessment, Safety Labs
Baseline	Day 0	Hospitalization confirmation, Randomization (KarXT vs. Placebo), First Dose
Interim	Acute Phase	Safety Labs, Efficacy Assessments (YMRS, CGI-BP), Monitoring for cholinergic AEs (nausea, vomiting, constipation)
End of Study	Follow-up	Final Vital Signs, Exit Interview, IP Accountability, Safety evaluation

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint 1. Change from baseline in Young Mania Rating Scale (YMRS) at Week 5. Evaluated via clinician-administered psychiatric rating scales.

7.2 Secondary & Exploratory Endpoints 1. Change from baseline in Clinical Global Impressions Bipolar (CGI-BP) at Week 5. 2. Occurrence of response at Week 5 assessed as the number of participants with a $\geq 50\%$ decrease from baseline in Young Mania Rating Scale (YMRS) score. 3. Occurrence of response at Week 5 assessed as the number of participants with a ≥ 1 change from baseline in CGI-BP. 4. Tolerability and safety parameters (incidence of short-term adverse events such as dry mouth, nausea, and constipation).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring during hospitalization for common AEs, especially GI-related cholinergic/anticholinergic effects. **Serious Adverse Events (SAEs):** Standard protocol requiring reporting within 24 hours of

site awareness. **Data Monitoring Committee (DMC):** Utilized by BMS for periodic safety evaluations across trial sites.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary efficacy analysis; Safety Set for all participants receiving at least one dose of the study drug. **Sample Size Justification:** Target enrollment of $N=424$ powered appropriately to detect statistical superiority of KarXT over placebo when added to standard mood stabilizers. **Handling of Missing Data:** Standard repeated measures (e.g., MMRM) or multiple imputation for longitudinal psychiatric endpoints.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Standardized clinical monitoring during the acute inpatient phase, followed by scheduled outpatient checks. **Audit Trail:** Secure eCRF systems deployed across all global centers for verified data capture and locking.

11. Study Identifiers & Governance

Primary ID: CN012-0046 **Posting ID:** 173442267910824732 **Registry Number:** NCT07140913 **Sponsor/Lead Organization:** Bristol Myers Squibb **Study Title:** A Study to Evaluate the Efficacy and Safety of Adjunctive KarXT for the Treatment of Mania, With or Without Mixed Features, in Participants With Bipolar-I Disorder Taking Lithium, Valproate, or Lamotrigine **Official Regulatory Title:** A Phase 3, Randomized, Double-blind, Placebo-controlled, Study to Evaluate the Efficacy and Safety of Adjunctive KarXT for the Treatment of Mania, With or Without Mixed Features, in Individuals With Bipolar-I Disorder Taking Lithium, Valproate, or Lamotrigine

12. Clinical Framework (Bipolar Disorder Specific)

Disease Areas: Mania, Bipolar Disorder **Primary Condition:** Bipolar I Disorder (Preferred UMLS Name), Acute Manic Episode (with or without mixed features) **Disease Name/Preferred Name:** Bipolar Disorder **Disease Definition:** A disorder of the brain that causes unusual shifts in mood, energy, activity levels and the ability to carry out day-to-day tasks. Often these moods range and shift from periods of elation and energized behavior to those of hopelessness and depression. **Semantic Type:** Mental or Behavioral Dysfunction **Diagnosis Threshold:** DSM-5-TR, confirmed by MINI

7.0.2 Standard, YMRS Total Score ≥ 18 , and CGI-BP ≥ 4
Medical Terminology Codes: * **MeSH Code:** D001714 * **HPO Code:** HP: 0007302 * **SNOMED ID:** 13746004 * **SNOMED Hierarchy:** 21071000119101 (Mood disorder of manic type / Mood disorder of manic type (disorder)); 46206005 (Mood disorder / Mood disorder (disorder)); 74732009 (Mental disorder / Mental disorder (disorder)); 64572001 (Disease / Disease (disorder)) **Medical Methodology:** Adjunctive Muscarinic Agonist Therapy (M1/M4) paired with standard mood stabilizers **Optimization Target:** Reduction in acute manic symptoms and CGI-BP illness severity score

13. Study Procedures & Methodology

Management Pathway: Inpatient hospitalization for acute symptom management **Education Protocol (HCP):** Site initiation and standardized rater training for YMRS and CGI-BP assessments **Education Protocol (Patient):** Informed consent and overview of expected mild/transient AEs (e.g., nausea, dry mouth) **Quality Audit Mechanism:** Blinded data reviews and inter-rater reliability checks for psychiatric scoring

14. Observation & Follow-Up Schedule

Total Duration: Acute hospitalization phase (up to 3 weeks) + Safety Follow-up phase (Primary endpoints assessed at Week 5) **Onsite Visit Cadence:** Daily/Weekly evaluations during the acute inpatient phase **Remote Monitoring Frequency:** Routine pharmacovigilance follow-up post-discharge **Checklist Review Interval:** Routine clinical assessments per protocol schedule

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]			Lead	
Statistician	[Name]	_____	[DD-MMM-YYYY]				